Fengwei Fried Rice · Stage 2 Test Bank: Answer Key

For the study guide "Hot Wok, Smart Chips" · Hardware, Networks & Behaviourist AI · November 2026

Part A · True or False (false statements are corrected)

1. True. One multiply plus one add = 2 FLOPs. [R1 · 1.1 · R]

2. False. Correct: the CPU is the most flexible chip; the TPU is the least flexible but the most efficient at matrix work. [R1 · 1.2 · U]

3. False. Correct: Amdahl's law gives 1 ÷ (0.1 + 0.9/100) ≈ 9.2×; the limit is 10×, set by the serial 10%. [R1 · 1.3 · A]

4. True. Arithmetic intensity is about 1 FLOP/byte, far below a ridge point of about 20 FLOP/byte. [R1 · 1.4 · U]

5. False. Correct: the speed-up is much SMALLER at size 256 (only a few times), because most GPU cores sit idle and launch overhead dominates; at 4,096 the GPU is roughly 50-100× faster. [R1 · Lab 1 · A]

6. True. Like boxes inside boxes: application data → TCP/UDP header → IP header → link header. [R2 · 2.1 · U]

7. False. Correct: UDP gives no delivery or ordering guarantee; TCP provides acknowledgements, retransmission and ordering. [R2 · 2.2 · U]

8. False. Correct: bandwidth is how much data per second; latency is how long one message takes to arrive. [R2 · 2.4 · U]

9. True. Traffic = $2(N - 1)/N \times S$, which tends to 2S for large N. [R2 · 2.4 · A]

10. True. One way = 6 ms = 6,000 µs; 6,000 ÷ 5 = 1,200 km. Real paths are slower, so the server is usually much closer. [R2 · Lab 2 · A]

11. True. $0.9^3 \times 10 = 7.29$. [R3 · 3.2 · A]

12. False. Correct: γ close to 0 makes the agent short-sighted (only the next reward counts); γ close to 1 is far-sighted. [R3 · 3.2 · U]

13. True. Example: the boat-racing agent circling to hit respawning targets instead of finishing the race. [R3 · 3.3 · U]

14. True. It never tries other options, so it never learns that a better one exists (0% best arm in the bandit simulation). [R3 · 3.4 · U]

15. False. Correct: with epsilon = 0 the agent never explores; it repeats the first route that worked, some squares are never tried, and it can get stuck on a poor route. [R3 · Lab 3 · U]

16. False. Correct: a camera measures light and colour; depth must be inferred (stereo, motion, learning). LiDAR, radar and ultrasonic measure distance directly. [R4 · 4.1 · R]

17. True. $d = c \times t \div 2 = 3 \times 10^{8} \times 4 \times 10^{-7} \div 2 = 60$ m. [R4 · 4.1 · A]

18. True. The frequency shift of the echo gives the relative speed in one measurement. [R4 · 4.1 · U]

19. False. Correct: correlated failures happen together; in the guide's example both-fail is 5% instead of 0.25% for independent sensors. [R4 · 4.4 · U]

20. False. Correct: the reading jitters around 0 (about ±0.05 m/s²) and is never exactly zero; that jitter is sensor noise σ. [R4 · Lab 4 · U]

21. True. Unfused: 3 × (read 2 MB + write 2 MB) = 12 MB. Fused: one read + one write = 4 MB. [R5 · 5.1 · A]

22. False. Correct: in data parallelism every chip holds a FULL copy of the model and processes a different slice of the batch; splitting the weights is model parallelism. [R5 · 5.2 · U]

23. True. PUE = total facility energy ÷ IT energy; 1.0 means zero overhead for cooling and power conversion. [R5 · 5.4 · U]

24. True. 43,800 ÷ 4,096 ≈ 10.7 h, which is why large jobs need checkpoints. [R5 · 5.4 · A]

25. True. The first call traces and compiles the function (about 0.1-1 s); later calls reuse the compiled code and take milliseconds. [R5 · Lab 5 · U]

Part B · Multiple choice

26. A) 1.0. $2 \times 1 + (-1) \times 4 + 0.5 \times 6 = 2 - 4 + 3 = 1.0$. [R1 · 1.1 · A]

27. C) NPU. An NPU gives the most work per watt within a few watts; GPUs and TPU pods need far more power. [R1 · 1.2 · U]

28. D) [17, 39]. $1 \cdot 5 + 2 \cdot 6 = 17$; $3 \cdot 5 + 4 \cdot 6 = 39$. [R1 · 1.2 · A]

29. B) 20 FLOP/byte. $10 \times 10^{12} \div (500 \times 10^{9}) = 20$ FLOP/byte; below 20 the chip is memory-bound. [R1 · 1.4 · A]

30. D) 5.5 TFLOP/s. TFLOP/s = 2 × MACs ÷ time = $2 \times 6.87 \times 10^{11} \div 0.25 \approx 5.5 \times 10^{12}$. 2.7 forgets the factor 2; 11 doubles it again. [R1 · Lab 1 · A]

31. A, B, D) Acknowledgement of delivered data / In-order delivery / Retransmission of lost packets. TCP adds acknowledgements, ordering and retransmission. Both TCP and UDP use port numbers, so C is wrong. [R2 · 2.2 · U]

32. A) QUIC. QUIC (RFC 9000) runs over UDP and combines transport and TLS handshakes. [R2 · 2.2 · U]

33. B) On or near the machine (edge). Only the edge fits a 15 ms budget; a far cloud round trip alone can exceed it. [R2 · 2.3 · U]

34. C) The slowest small fraction of requests. For example the 99th-percentile delay; one slow server can delay a whole request. [R2 · 2.4 · U]

35. A) Putting computing close to the base station. Operator servers near the base station shorten the round trip. [R2 · 2.5 · U]

36. C) 0.81. V(3) = 1, V(2) = 0.9 × 1 = 0.9, V(1) = 0.9 × 0.9 = 0.81. [R3 · 3.2 · A]

37. D) 0.325. Target = 1 + 0.9 × 0.5 = 1.45; TD error = 1.25; new Q = 0.2 + 0.1 × 1.25 = 0.325. [R3 · 3.3 · A]

38. A, C) +speed every step / −energy used every step. A and C give feedback every step. B and D are sparse: they arrive only at the end. [R3 · 3.3 · U]

39. B) 0.77. 0.9 × 0.8 + 0.1 × (0.2 + 0.5 + 0.8) ÷ 3 = 0.72 + 0.05 = 0.77. Careful: if exploration chose only the OTHER two arms the answer would be 0.9 × 0.8 + 0.1 × 0.35 = 0.755. [R3 · 3.4 · A]

40. D) Training on many varied simulated conditions. If the policy works across many random variations, the real world looks like one more variation. [R3 · 3.5 · U]

41. A, C, D) LiDAR / Radar / Ultrasonic. LiDAR (light time of flight), radar (radio echo) and ultrasonic (sound echo) measure range; a single camera must infer depth. [R4 · 4.1 · U]

42. A) 8.33 m. 100 km/h = 27.78 m/s; 27.78 × 0.3 ≈ 8.33 m before the car even reacts. [R4 · 4.2 · A]

43. B) 50.12 m. Weights $1/\sigma^2$: 4 and 0.25. (4 × 50 + 0.25 × 52) ÷ 4.25 = 213 ÷ 4.25 ≈ 50.12 m: the precise radar dominates. [R4 · 4.3 · A]

44. C) 50 m. Z = f × B ÷ d = 1,000 × 0.3 ÷ 6 = 50 m. [R4 · 4.3 · A]

45. A) Level 2. Level 2 = partial automation; the human remains the driver. [R4 · 4.4 · U]

46. C) 112 GB. $7 \times 10^9 \times 16$ B = 112 GB: more than one chip's memory, so the state must be split. [R5 · 5.2 · A]

47. D) 143 tokens/s. Every token reads all 7 GB: 7 GB ÷ 1 TB/s = 7 ms per token → about 143 tokens/s. [R5 · 5.3 · A]

48. B) 24 hops. Torus: 3 × (16 ÷ 2) = 24 hops. 45 is the mesh without wraparound. [R5 · 5.5 · A]

49. A, B, D) Fusing operations to cut memory traffic / Overlapping communication with computation / Using a larger batch per chip. A, B and D keep the compute units busier. C leaves the chips idle, so MFU falls. [R5 · 5.4 · U]

50. D) 8×. Matrix multiply = $2n^3$ FLOPs; doubling n gives $2^3 = 8$×. For large sizes the measured time grows close to 8× too. [R5 · Lab 5 · A]

Part C · Open questions (sample answers, key points and marking guide)

Each open question is worth 5 marks. The key points below are a guide, not the only correct answer: give credit to any valid alternative that is correct and well reasoned.

51. Weights 512 × 2,048 = 1,048,576 → 2.1 MB; FLOPs per input 2,097,152. Batch 1 ≈ 1.0 FLOP/byte < 20 → memory-bound, ≈ 0.5 TFLOP/s (5%). Batch 32 ≈ 29.7 FLOP/byte > 20 → compute-bound, ≈ 10 TFLOP/s. Batching reuses each fetched weight for many inputs, raising FLOPs per byte. Marks: intensities (2), classification and TFLOP/s (1), explanation (2). Accept small differences from rounding or from ignoring activation bytes. [R1 · 1.4 · A]

52. Advantages: privacy (face data stays in the Secure Enclave), low latency, works offline, energy-efficient NPU. Limitations: limited compute and memory, unsupported operations fall back to the CPU, models must be small or quantised. Large chatbot: usually cloud or hybrid, because its weights exceed phone memory (or a small on-device model + cloud). Marks: 1 per valid point (4), justified decision (1). Accept other valid points, e.g. updates or battery life. [R1 · 1.5 · E]

53. Strong answers choose an on-premises edge GPU server or hybrid: data stays inside the hospital (privacy and law), LAN latency is low, X-ray images are large (bandwidth), the cloud can still be used for training or overflow. Cost: buying a server vs renting. Marks: recommendation (1), four justified factors (4). Accept a well-argued cloud answer that addresses privacy (encryption, anonymisation, contracts) and the 2-second budget. [R2 · 2.3 · E]

54. Traffic = 2 × 255/256 × 14 ≈ 27.9 GB; time ≈ 0.28 s; efficiency = 8.2 ÷ (8.2 + 0.28) ≈ 96.7%. Improvements: overlap communication with compute, larger batch per chip (more compute per byte), gradient compression, hierarchical collectives. Marks: 3 calculations, 2 for one improvement with a correct reason. [R2 · 2.4 · A]

55. Naive: +1 per item put in the bin → the robot tips the bin and re-collects. Better: reward the reduction of rubbish on the floor (or clean area), penalties for knocking things

over and for energy. Dense if given every step, sparse if only when the room is clean. Comparison: the naive one rewards activity, the better one rewards the real goal. Marks: hack (2), improved reward (2), sparse/dense with comparison (1). Accept other plausible hacks. [R3 · 3.3 · C]

56. Failure: epsilon = 0 means no exploration; the agent exploits the first route that reached the goal, so unvisited squares are never updated and a shorter route is never found. Fix: epsilon about 0.1-0.2 (or decaying epsilon). Epsilon = 0.5: values are learned everywhere but the path looks messy and hits the −1 squares more often. Bandit link: the greedy agent found the best arm 0% of the time. ε-greedy explores at random at a fixed rate; UCB adds a bonus $\sqrt{(2 \ln t / n)}$ to little-tried actions and explores where it is uncertain. Marks: diagnosis (2), fix (1), ε = 0.5 prediction (1), comparison (1). Accept other reasonable fixes such as optimistic initial values. [R3 · Lab 3 · An]

57. 13.89 m/s; blind distance 2.08 m; braking $13.89^2 \div 14 \approx 13.78$ m; total ≈ 15.9 m. Detected at 15 m: 12.92 m left to brake, $v = \sqrt{(13.89^2 - 2 \times 7 \times 12.92)} \approx 3.5$ m/s at impact. Lesson: early, reliable detection (range) and low latency decide whether braking works; both must be in the design budget. Marks: 2 + 2 + 1. [R4 · 4.4 · A]

58. Speeds: 95, 250, 133, 67, 182 m/s. Mean t = 0.17 s → v ≈ 118 m/s (average the times, not the speeds). True delay is only 20 ÷ 343 ≈ 0.058 s, shorter than human reaction time (about 0.2 s), so reaction errors are as big as the signal. Averaging n trials shrinks random noise by about $\sqrt{n}$ (sensor fusion idea, 4.3) but cannot remove a systematic bias, e.g. reacting faster to sound than to sight. Improvements: a longer distance (100 m → 0.29 s), many more trials, or phyphox Acoustic Stopwatch on two phones (no human reaction). Marks: calculations (2), noise explanation (1), averaging vs bias (1), improvement (1). [R4 · Lab 4 · An]

59. $6 \times 7 \times 10^9 \times 1.4 \times 10^{12} = 5.88 \times 10^{22}$ FLOPs; $256 \times 200 \times 10^{12} \times 0.4 = 2.05 \times 10^{16}$ FLOP/s → ≈ 2.87×10^6 s ≈ 33.2 days; ≈ 204,167 chip-hours ≈ $408,333; 204,167 h × 0.4 kW × 1.1 ≈ 89.8 MWh. Raising MFU (fusion, better batching, overlap) cuts time, cost and energy in proportion. Marks: 4 values, 1 for the change with reason. Accept "more chips" only if the answer notes that cost and energy stay about the same. [R5 · 5.4 · A]

60. First call: tracing + XLA compilation. FLOPs = $2n^3$: 1,024 → 2.15 GFLOP ÷ 0.0021 s ≈ 1.0 TFLOP/s (≈ 13% MFU); 4,096 → 137 GFLOP ÷ 0.034 ≈ 4.0 TFLOP/s (≈ 50%); 8,192 → 1.10 TFLOP ÷ 0.27 ≈ 4.1 TFLOP/s (≈ 50%). 1,024 → 4,096 is 64× the FLOPs but only ≈ 16× the time, because at small sizes fixed overhead (launch, transfer) dominates and the GPU is not full. 4,096 → 8,192: 0.27 ÷ 0.034 ≈ 7.9 ≈ 8×, both sizes are compute-bound. Prediction for 16,384: about 8 × 0.27 ≈ 2.2 s, if memory allows. Marks: compilation (1), TFLOP/s and MFU (2), ratio explanation (1), prediction (1). Accept answers that also mention timing noise or GPU clock changes. [R5 · Lab 5 · P]